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import re

# ── TASK 4: Basic Rule-Based Chatbot ──────────────────────────────────────────
# Key concepts: if-elif, functions, loops, input/output

# ── Response rules: (pattern, reply) ──────────────────────────────────────────
RULES = [
    # Greetings
    (r"\b(hello|hi|hey|howdy|hiya)\b",                 "Hey there! 👋 How can I help you today?"),
    (r"\bgood (morning|afternoon|evening)\b",    "Good {1}! Hope you're having a great day! 😊"),

    # Asking how the bot is
    (r"how are you(\?)?",                           "I'm doing great, thanks for asking! What about you?"),
    (r"(i am|i'm|im) (good|fine|great|okay|ok)", "Glad to hear that! 😄 Anything I can help you with?"),
    (r"(i am|i'm|im) (bad|sad|not good|terrible|awful)", "I'm sorry to hear
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that 😔. Want to talk about it?"),

    # Name
    (r"what('s| is) your
name",                  "I'm CodeBot — your
friendly rule-based assistant! 🤖"),
    (r"(my name is|i am|i'm|im)
(\w+)",           "Nice to meet you, {2}!
😊"),

    # Age
    (r"how old are
you",                        "I was just
created, so I guess I'm 0 years old! 😄"),
    (r"what('s| is) your
age",                   "Age is just a
number — especially for bots!"),

    # Capabilities
    (r"what can you
do",                         "I can chat
with you, answer simple questions, and keep
you company! 🗣️"),
    (r"are you (a )?
(bot|robot|ai|chatbot)",     "Yes! I'm a
simple rule-based chatbot built in Python.
🤖"),
    (r"are you
human",                           "Nope!
I'm 100% code. But I try to be helpful
anyway. 😄"),

    # Jokes
    (r"tell me a
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joke",                          "Why do
programmers prefer dark mode? Because light
attracts bugs! 🐛😄"),
    (r"another
joke",                            "Why was
the computer cold? Because it left its
Windows open! 💻"),

    # Time / Date
    (r"what (time|day|date) is
it",              "I don't have a clock, but
Python's `datetime` module does! ⏰"),

    # Python / coding
    (r"(do you|can you)
(know|code|program)",    "Yes! I'm written
in Python. Coding is my whole existence!
🐍"),
    (r"(what is|what's)
python",                 "Python is a
powerful, beginner-friendly programming
language. You're using it right now! 🐍"),

    # Thank you
    (r"\b(thanks|thank
you|thx|ty)\b",           "You're welcome!
😊 Happy to help!"),

    # Goodbye
    (r"\b(bye|goodbye|see
you|cya|quit|exit)\b", "GOODBYE"),

    # Fallback
]
```

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BOT_NAME = "CodeBot"
USER_NAME = "You"


def get_response(user_input: str) -> str:
    """Match user input against rules and
return an appropriate response."""
    text = user_input.lower().strip()

    for pattern, response in RULES:
        match = re.search(pattern, text)
        if match:
            if response == "GOODBYE":
                return "GOODBYE"
            # Fill in matched groups like
{1}, {2}
            try:
                groups = match.groups()
                for i, g in
enumerate(groups, 1):
                    if g:
                        response =
response.replace(f"{{{i}}}", g)
            except Exception:
                pass
            return response

    # Default fallback
    fallbacks = [
        "Hmm, I'm not sure I understand.
Could you rephrase that? 🤔",
        "Interesting! Tell me more. 🙂",
        "I'm still learning! Try asking me
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something else.",
        "That's a bit beyond my rule-book.
Ask me something simpler! 😊",
    ]
    import random
    return random.choice(fallbacks)


def run_chatbot():
    print("\n" + "=" * 50)
    print(f"       🤖 Welcome to
{BOT_NAME}!")
    print("=" * 50)
    print("  Type 'bye' or 'exit' to
quit.\n")

    while True:
        user_input = input(f"  {USER_NAME}:
").strip()

        if not user_input:
            print(f"  {BOT_NAME}: Please
say something! I'm listening. 👂\n")
            continue

        response = get_response(user_input)

        if response == "GOODBYE":
            print(f"  {BOT_NAME}: Goodbye!
It was great chatting with you. Take care!
👋\n")
            break

        print(f"  {BOT_NAME}:
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{response}\n")



if __name__ == "__main__":
    run_chatbot()
```